University of Central Punjab

Faculty of Information Technology

Object Oriented Programming

Fall 2025

Lab 02

Topic
Functions
Pass-by-Value, Pass-by-Reference
Function Overloading, Default Arguments
* Dynamic Memory Allocation, Dangling Pointers, and Memory Leakage

Objective
The basic purpose of this lab is to revise some preliminary concepts of C++ that has been covered in the course of Introduction to Computing and Programming Fundamentals.
Its objective is to recall previously learned basic concepts like revision of arrays and functions

CLO
Apply object-oriented programming principles to implement real-world problems.

Instructions:

- Indent your code.
- Comment your code.
- Use meaningful variable names.
- Plan your code carefully on a piece of paper before you implement it.
- Name of the program should be same as the task name. i.e. the first program should be Task_1.cpp
- **void main() is not allowed. Use int main()**
- **You are not allowed to use any built-in functions**
- **You are required to follow the naming conventions as follow:**
    - **Variables:** firstName; (no underscores allowed)
    - **Function:** getName(); (no underscores allowed)
    - **ClassName:** BankAccount (no underscores allowed)

Students are required to complete the following tasks in lab timings.

Task 1:

Write a C++ function to swap two integers. Implement this function in two ways:

- One using pass-by-value
- One using pass-by-reference

void swapByValue(int a, int b);

void swapByReference(int &a, int &b);

In main function, demonstrate each version of the **'Swap** function and print the results.

Task 2:

Write an overloaded function named **‘average’** that calculates:

- Average of two **integers**.
- Average of two **doubles**.

You can use return type for functions of your own choice (void, int, double).

void average(int a, int b);

void average(double a, double b);

In the main function, demonstrate each version of the **‘average’** function and print the results.

Task 3:

Write an overloaded function named **‘Area’** that will calculate:

- Area of a **Square**.
- Area of a **Triangle**.
- Area of a **Circle**.

You can use return type for functions, of your own choice (void, double, int).

void Area(int length);

void Area(int length, int width);

void Area(double radius);

In the main function, demonstrate each version of the **‘Area’** function and print the results.

Task 4:

Write a C++ program that defines a function calculate that computes the area of a rectangle.
The function should accept two arguments:

- length
- width

The width argument should have a default value of 5.

int area1 = calculate(10); // Uses default width = 5
int area2 = calculate(10, 3); // Uses provided width = 3

Demonstrate how the function behaves when the width is provided and when it defaults to 5.

Task5:

Write a C++ program that:

- Dynamically allocates memory for an array of 5 integers.
- Assigns values to the array.
- Demonstrates the problem of dangling pointers by deleting the memory and trying to access the deleted pointer.
- Shows an example of memory leakage where memory is allocated but not properly deallocated before the program exits.

Task 6:

Now create a menu based program where user should select the function that needs to be tested from the above functions.

For example:

- Press 1 for PassByValue and PassByRef Function
- Press 2 for Overloaded Average Function
- Press 3 for Overloaded Area Function
- Press 4 for Default Argument **Calculate** Function.
- Press 5 for Testing Dynamic memory allocation, dangling pointers, and memory leak.

- Press 0 for exit.